Introduction to SQL

Module 7: Inserting, Updating, and Deleting Data

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CRUD

In SQL: `INSERT` `SELECT` `UPDATE` `DELETE`

INSERT a new row INTO a table

SQL

```sql
INSERT INTO table (column1, column2)
   VALUES (value1, value2);
```

INSERT a row INTO a table

Employee table

EmployeeID	FirstName	LastName	Email	HireDate	Role	IsFullTime	Department
592	Isaac	Morrison	isaac.morrison@example.com	2022-03-15	Manager	TRUE	Finance
593	Leah	Ramirez	leah.ramirez@example.com	2021-06-28	Developer	TRUE	Engineering
594	Gabriel	Gomez	gabriel.gomez@example.com	2023-01-10	Designer	TRUE	
595	Ella	Kim	ella.kim@example.com	2022-09-05	Marketing Specialist	FALSE	Marketing
596	Connor	Ferguson	c.ferguson@example.com	2021-11-20	Senior Developer	TRUE	Engineering
597	Levi	Nguyen	levi.nguyen@example.com	2023-04-02	HR Manager	TRUE	
598	Maya	Howard	maya.howard@example.com	2022-07-18	Product Manager	TRUE	
599	Blake	Patel	blake.patel@example.com	2023-02-14	Data Analyst	TRUE	General
600	Harper	Reid	harper.reid@example.com	2021-08-09	Software Engineer	TRUE	Engineering
601	Hudson	Cooper	hudson.cooper@example.com	2022-05-25	Graphic Designer	FALSE	General
602	Jane	Doe	jd@example.com	2024-01-01	Developer	TRUE	General

Column Definitions:

```
EmployeeID INT PRIMARY KEY AUTO_INCREMENT
FirstName VARCHAR(50) NOT NULL
LastName VARCHAR(100) NOT NULL
HireDate DATE NOT NULL
Role VARCHAR(255) NOT NULL
IsFullTime BOOLEAN NOT NULL
Department VARCHAR(255) DEFAULT 'General'
```

```
INSERT INTO Employee (FirstName, LastName, Email, HireDate, Role, IsFullTime)
       VALUES ('Jane','Doe','jd@example.com','2024-01-01','Developer',TRUE);
```

INSERT multiple rows INTO a table

Employee table

EmployeeID	FirstName	LastName	Email	HireDate	Role	IsFullTime	Department
592	Isaac	Morrison	isaac.morrison@example.com	2022-03-15	Manager	TRUE	Finance
593	Leah	Ramirez	leah.ramirez@example.com	2021-06-28	Developer	TRUE	Engineering
594	Gabriel	Gomez	gabriel.gomez@example.com	2023-01-10	Designer	TRUE	
595	Ella	Kim	ella.kim@example.com	2022-09-05	Marketing Specialist	FALSE	Marketing
596	Connor	Ferguson	c.ferguson@example.com	2021-11-20	Senior Developer	TRUE	Engineering
597	Levi	Nguyen	levi.nguyen@example.com	2023-04-02	HR Manager	TRUE	
598	Maya	Howard	maya.howard@example.com	2022-07-18	Product Manager	TRUE	
599	Blake	Patel	blake.patel@example.com	2023-02-14	Data Analyst	TRUE	General
600	Harper	Reid	harper.reid@example.com	2021-08-09	Software Engineer	TRUE	Engineering
601	Hudson	Cooper	hudson.cooper@example.com	2022-05-25	Graphic Designer	FALSE	General
602	Jane	Doe	jd@example.com	2024-01-01	Developer	TRUE	General
603	Clair	Flores	cf@example.com	2024-01-02	Designer	TRUE	General
604	Eliana	Brooks	ebrooks@example.com	2024-01-03	Developer	TRUE	General
602	Nathan	Cruz	ncruz@example.com	2024-01-04	Social Media Intern	FALSE	General

Column Definitions:

```
EmployeeID INT PRIMARY KEY AUTO_INCREMENT
FirstName VARCHAR(50) NOT NULL
LastName VARCHAR(100) NOT NULL
HireDate DATE NOT NULL
Role VARCHAR(255) NOT NULL
IsFullTime BOOLEAN NOT NULL
Department VARCHAR(255) DEFAULT 'General'
```

```
INSERT INTO Employee (FirstName, LastName, Email, HireDate, Role, IsFullTime)
   VALUES ('Claire','Flores','cf@example.com','2024-01-02','Designer',TRUE),
          ('Eliana','Brooks','ebrooks@example.com','2024-01-03','Developer',TRUE),
          ('Nathan','Cruz','ncruz@example.com','2024-01-04','Social Media Intern',False);
```

clip 2: update

Using UPDATE to change values in a row

SELECT statement

```
SELECT  (columns)
FROM  (table)
WHERE  (condition)
```

UPDATE statement

```
UPDATE  (table)
SET  (column = value)
WHERE  (condition)
```

Using UPDATE to change values in a row

SQL

```sql
UPDATE Employee
SET Status = 'Retired'
WHERE EmployeeID = 6543;
```

Employee table

EmployeeID	FirstName	LastName	Email	DepartmentID	Status	DateHired	Role	Salary
6541	Lyla	Peterson	lyla.peterson@example.com	1	Retired	2023-01-05	Marketing Lead	95000
6542	Owen	Brooks	owen.brooks@example.com	2	Full-time	2021-07-22	Sales Representative	70000
6543	Isaac	Morrison	isaac.morrison@example.com	5	Retired	2022-03-15	Comptroller	75000
6544	Lean	Ramirez	lean.ramirez@example.com	3	Part-time	2021-06-28	Research Assistant	90000
6545	Gabriel	Gomez	gabriel.gomez@example.com	2	Parttime	2023-01-10	Sales Manager	80000
6546	Ella	Kim	ella.kim@example.com	2	Full-time	2022-09-05	Relationship Manager	70000
6547	Connor	Ferguson	c.ferguson@example.com	1	Full-time	2021-11-20	Lead Generation	50000
6548	Levi	Nguyen	levi.nguyen@example.com	8	Full-time	2023-04-02	Operational Assistant	85000
6549	Maya	Howard	maya.howard@example.com	3	Retired	2022-07-18	Product Manager	95000
6550	Blake	Patel	blake.patel@example.com	1	Full-time	2023-02-14	Marketing Specialist	75000
6551	Harper	Reid	harper.reid@example.com	6	Resigned	2021-08-09	Software Engineer	85000
6552	Hudson	Cooper	hudson.cooper@example.com	1	Full-time	2022-05-25	Graphic Designer	70000
6553	Grace	Evans	grace.evans@example.com	5	Sabbatical	2022-12-10	Finance Analyst	80000
6554	Luke	Stewart	luke.stewart@example.com	5	Full-time	2023-03-18	Bookkeeper	75000
6555	Claire	Flores	claire.flores@example.com	5	Full-time	2021-09-30	Tax Specialist	110000
6556	Aiden	Ward	aiden.ward@example.com	6	Intern	2022-04-12	UX Designer	85000

```
> 1 row affected
```

Using UPDATE to change values in a row

SQL

```sql
UPDATE Employee
SET Status = 'Retired', Salary = 0, DepartmentID = NULL
WHERE EmployeeID = 6543;
```

The DBMS will validate data types

SQL

```
UPDATE Product
    SET Quantity = 'ABCD1234'
    WHERE ProductID = 11388;
```

Product table

defined as integer

ProductID	ProductName	Description	Category	UnitWeight	ListPrice	Quantity
11381	Laptop	High-performance laptop with SSD storage	Electronics	2.5	1200	50
11382	Smartphone	Latest smartphone with advanced camera features	Electronics	0.2	800	84
11383	Headphones	Noise-canceling headphones with Bluetooth connectivity	Electronics	0.3	99	207
11384	T-shirt	Cotton t-shirt available in various colors	Clothing	0.4	20	300
11385	Jeans	Classic denim jeans for everyday wear	Clothing	0.8	40	150
11386	Running Shoes	Lightweight running shoes for athletes	Footwear	0.6	80	54
11387	Backpack	Durable backpack with multiple compartments	Accessories	1.2	50	150
11388	Watch	Stylish wristwatch with leather strap	Accessories	0.1	115	120
11389	Bluetooth Speaker	Portable speaker with long battery life	Electronics	1.0	70	80
11390	Desk Lamp	LED desk lamp with adjustable brightness	Home & Office	0.5	105	182
11391	Notebook	Spiral-bound notebook for writing and sketching	Stationery	0.2	5	5
11392	Water Bottle	Stainless steel water bottle for hydration on-the-go	Accessories	0.4	15	12

```
Error: Incorrect integer value: 'ABCD1234' for column 'Quantity'
```

clip 3: delete

Using DELETE to delete entire rows

SQL

```
DELETE FROM Product
    WHERE ProductID = 11382;
```

Product table

ProductID	ProductName	Description	Category	UnitWeight	ListPrice	Quantity
11381	Laptop	High-performance laptop with SSD storage	Electronics	2.5	1200	50
11382	Smartphone	Latest smartphone with advanced camera features	Electronics	0.2	800	84
~~11383~~	~~Headphones~~	~~Noise-canceling headphones with Bluetooth connectivity~~	~~Electronics~~	~~0.3~~	~~99~~	~~287~~
11384	T-shirt	Cotton t-shirt available in various colors	Clothing	0.4	20	300
11385	Jeans	Classic denim jeans for everyday wear	Clothing	0.8	40	150
11386	Running Shoes	Lightweight running shoes for athletes	Footwear	0.6	80	54
11387	Backpack	Durable backpack with multiple compartments	Accessories	1.2	50	150
11388	Watch	Stylish wristwatch with leather strap	Accessories	0.1	115	120
11389	Bluetooth Speaker	Portable speaker with long battery life	Electronics	1.0	70	80
11390	Desk Lamp	LED desk lamp with adjustable brightness	Home & Office	0.5	105	182
11391	Notebook	Spiral-bound notebook for writing and sketching	Stationery	0.2	5	5
11392	Water Bottle	Stainless steel water bottle for hydration on-the-go	Accessories	0.4	15	12

```
> 1 row affected
```

Using DELETE to delete multiple rows

SQL

```
DELETE FROM Product
  WHERE ProductID IN (11381, 11382);
```

Product table

ProductID	ProductName	Description	Category	UnitWeight	ListPrice	Quantity
~~11381~~	~~Laptop~~	~~High-performance laptop with SSD storage~~	~~Electronics~~	~~2.5~~	~~1200~~	~~50~~
~~11382~~	~~Smartphone~~	~~Latest smartphone with advanced camera features~~	~~Electronics~~	~~0.2~~	~~800~~	~~94~~
11384	T-shirt	Cotton t-shirt available in various colors	Clothing	0.4	20	300
11385	Jeans	Classic denim jeans for everyday wear	Clothing	0.8	40	150
11386	Running Shoes	Lightweight running shoes for athletes	Footwear	0.6	80	54
11387	Backpack	Durable backpack with multiple compartments	Accessories	1.2	50	150
11388	Watch	Stylish wristwatch with leather strap	Accessories	0.1	115	120
11389	Bluetooth Speaker	Portable speaker with long battery life	Electronics	1.0	70	80
11390	Desk Lamp	LED desk lamp with adjustable brightness	Home & Office	0.5	105	182
11391	Notebook	Spiral-bound notebook for writing and sketching	Stationery	0.2	5	5
11392	Water Bottle	Stainless steel water bottle for hydration on-the-go	Accessories	0.4	15	12

```
> 2 rows affected
```

DELETE without WHERE is a recipe for disaster

SQL

```
DELETE FROM Employee
```

Employee table

EmployeeID	FirstName	LastName	Email	DepartmentID	Status	DateHired	Role	Salary

```
> 11384 rows affected
```

Best practice: DELETE with primary key

SQL

```sql
DELETE FROM Employee
   WHERE EmployeeID = 6547;
```

Note: some DBMS use a "safe mode", where all DELETE statements must contain a WHERE clause using a primary key.